

Data visualization – Process Book – Milestone 3



A Fishing Story

Four years of global fishing vessel activity. Explore who fishes where, which waters they enter, and the patterns that define our oceans.

4 years

OF VESSEL DATA

Global

OCEAN COVERAGE

60 months

OF OBSERVATIONS

ACCESS THE MAP →

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From Idea to Visualization

The project started from a dataset search. Among the many options we explored, the Global Fishing Watch AIS data immediately stood out for its originality, a globally aggregated, high-quality record of fishing vessel activity detected via satellite and AIS transponders. We quickly saw the geopolitical potential it carried: rather than simply showing *where* fishing happens, we could ask *under whose authority* it happens. This angle, sitting at the intersection of environmental data and international law, matched our personal interest in geopolitics and felt genuinely underexplored compared to existing work on the dataset.

To make this angle concrete, we introduced a second dataset: the Marine Regions EEZ shapefiles, which provide the sovereign maritime boundaries of every coastal nation. By spatially joining fishing activity with these boundaries, we could classify each data point into one of three categories ; domestic fishing (a vessel operating within its own country's EEZ), foreign incursion (a vessel fishing inside another country's EEZ), or high seas activity (international waters with no sovereign jurisdiction). This classification became the backbone of the entire narrative.

Development then followed a progressive path. We started by building a functional V1 with a basic map rendering fishing vessel positions, establishing the frontend infrastructure and data pipeline. From this foundation, we layered features one by one:

- A fully redesigned map, significantly more polished visually, with EEZ boundaries rendered directly on top of the fishing activity layer, each zone colored by country.
- A chronological slider at the bottom of the screen, allowing users to scrub through time and observe how fishing patterns evolve across years.
- Adaptive statistical charts linked to the current map view, updating in real time as the user pans and zooms, showing the distribution of fishing hours by country, the share of foreign incursions, and the evolution of activity over time in the visible area.
- A country filter, letting users isolate the vessels of a single nation and trace their activity across the globe.
- Narrative points of interest (POIs) placed on geopolitically significant maritime regions, which trigger a zoom and display contextual explanations when clicked.

The final product is considerably richer than what was planned at Milestone 2, both visually and interactively, but the core question “*who is fishing in whose waters?*” never changed from the very first conversation.

Technical Design

Dataset and Constraint

Our primary dataset, Global Fishing Watch's AIS Apparent Fishing Effort, in which we focus on the five most recent years (2020–2024), represents close to 2 billion records distributed across ~93 GB of CSV files. The data is organized as one CSV per day at 0.01° spatial resolution, with each row reporting the fishing hours and number of distinct vessels for a given combination of (date, grid cell, flag, gear type).

This volume immediately collides with a deployment constraint: the visualization must run without any backend. Hence, it can only rely on static files and in-browser computations. The challenge then becomes: how does a browser interactively explore an aggregate of 2 billion rows over a global map?

Data Processing

The pipeline is built around the idea that every byte the browser will ever need is precomputed and shipped as a static file. It is implemented as a series of notebooks producing Parquet outputs. We first convert all CSVs to Parquet format, aggregating on axes that we do not consider, such as gear type. This alone shrinks the data by a factor of seven, from 93 GB to ~14 GB. We then perform a one-time point-in-polygon join against Marine Regions' EEZ shapefile, producing a 13MB file that maps every 0.1° cell to its sovereign zone: the geopolitical layer of the visualization. Finally, dedicated aggregates are precomputed for views that don't need daily granularity: a single 70 MB monthly time series file, and one ~10 MB file per narrative zone holding a full year of locally aggregated, EEZ-tagged activity.

Deployment

The processed files (~8 GB across roughly 1,830 Parquet files) are published as a public dataset on Hugging Face Hub, which we use purely as a static, CDN-backed file host. The frontend then queries these files directly from the browser using DuckDB WASM, an analytical SQL engine compiled to WebAssembly.

The key design decision was how the daily fishing data is split. Ideally, DuckDB would fetch only the Parquet footer and the relevant data via HTTP range requests, sparing the user from downloading entire monthly files (~100–250 MB each). In practice, we discovered that Hugging Face's CDN redirects break byte-range sub-requests, forcing DuckDB to fall back to full-file downloads. We addressed this by splitting the data into 1,825 per-day files of 3–5 MB each. Loading the data for a single day then costs one small HTTPS GET, cached by the browser thereafter.

From Sketches to Final Product

No formal sketches were produced at Milestone 1, the team moved directly to implementation. We very quickly had a working map that already matched what would become the Milestone 2 prototype: a full-screen interactive world map with fishing activity rendered as a heatmap, with panning and zooming fully functional. At that stage, the points of interest and the chronological slider were not yet integrated, and the statistical charts were still under development.

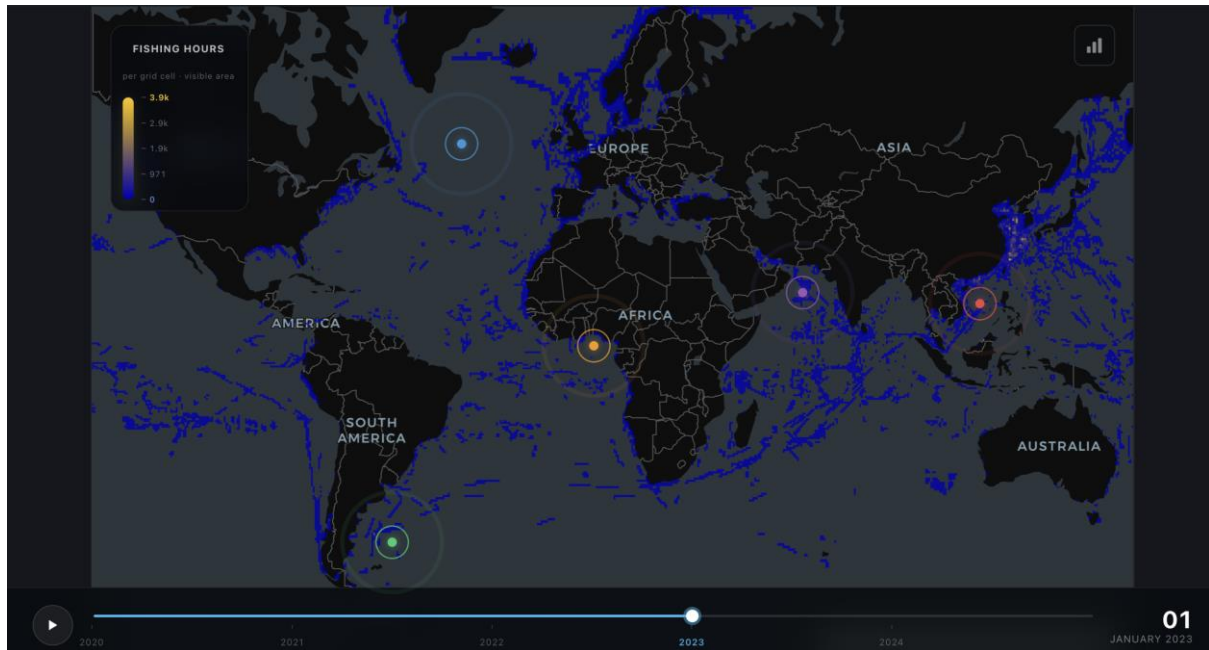


Figure 1 : Visualization at milestone 2

The final version departs from this starting point in two significant ways.

Visually, the map was entirely redesigned. We moved away from the dark neon aesthetic toward a cleaner, atlas-style cartographic base, light backgrounds, readable country labels, and a more restrained color palette for the fishing activity layer. This makes the visualization considerably more readable, especially once additional layers are overlaid on top of the map.

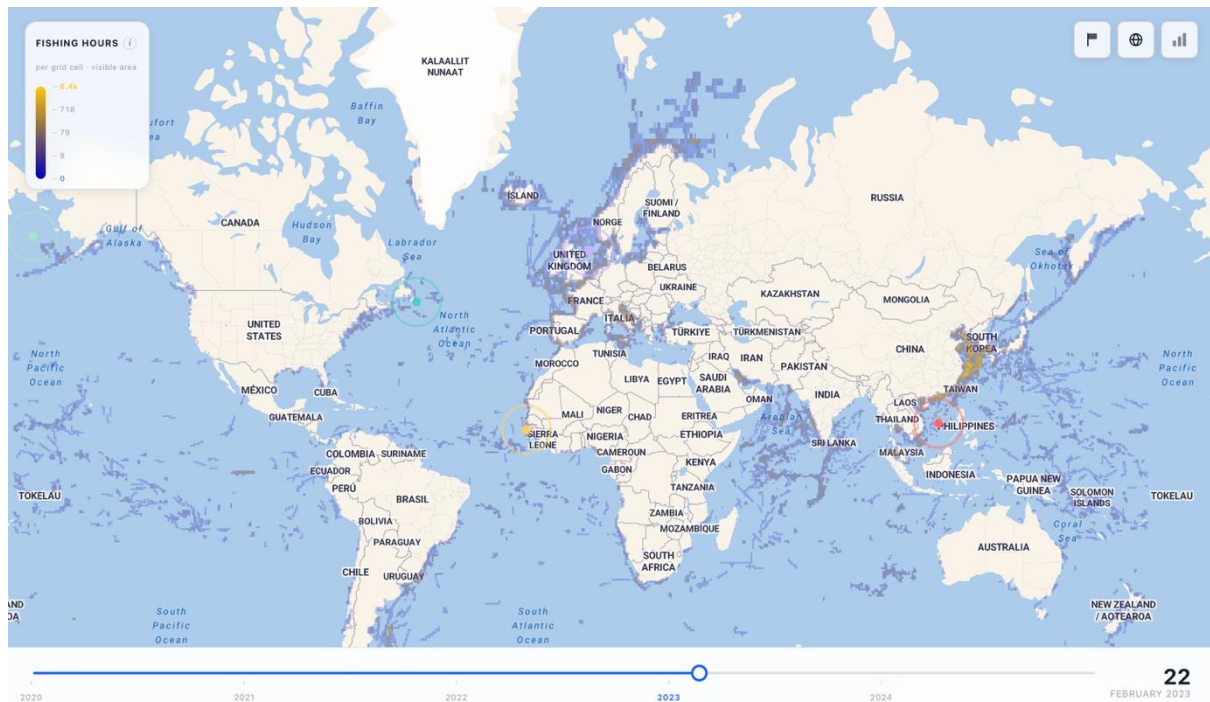


Figure 2 : Final general visualization

Functionally, four major features complete the interface:

1/ The statistical panel, opened via a button in the top-right corner, displays three adaptive charts that update in real time as the user pans and zooms.

The first, *Fishing Effort by Country*, is a horizontal bar chart showing the yearly share of fishing hours per country within the visible area. It gives an immediate sense of which nations dominate fishing activity in any given region, a simple but powerful way to reveal concentration and inequality across the globe.

The second, *Foreign EEZ Fishing*, compares for the top 5 countries in the zone the proportion of their fishing effort spent in foreign EEZs versus their home waters or international seas. This is the most direct visual expression of our geopolitical angle: it makes foreign incursions quantifiable at a glance, and exposes which nations are most dependent on fishing outside their own sovereign waters and international seas.

The third, *Fishing Activity Over Time*, is a monthly bar chart showing vessel presence in the visible area across the selected year. By revealing seasonal rhythms and year-to-year trends, it adds a temporal dimension to the spatial exploration, making it possible to spot anomalies or shifts in fishing pressure over time.

All three charts are rendered as custom SVG without any external charting library, giving full control over layout, animation, and styling.

2/ **The points of interest** are narrative anchors placed on geopolitically significant maritime regions. When clicked, they trigger an automatic zoom into the relevant area and open a side panel combining a contextual text analysis with the adaptive statistical charts for that specific zone turning a passive map into a guided story.

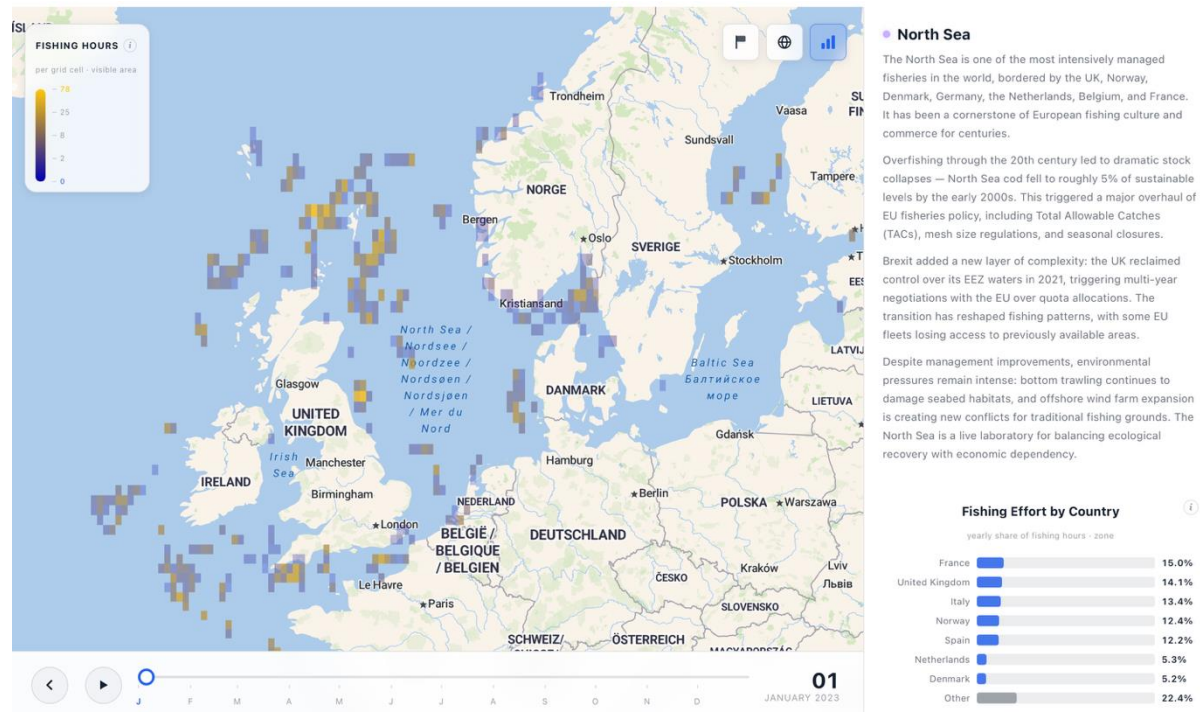


Figure 3 : “North Sea” point of interest visualization

3/ **The country filter** lets users isolate vessels by flag state, allowing direct visual comparison of how different nations distribute their fishing effort across domestic waters, foreign EEZs, and the high seas.

4/ **The EEZ layer** can be toggled on via a dedicated button, rendering each country's sovereign maritime boundaries directly on the map. This is the visual backbone of our geopolitical angle, it makes foreign incursions immediately readable by eye, without any additional explanation.

Let's study a **use case** to see the full potential of *A Fishing Story*.

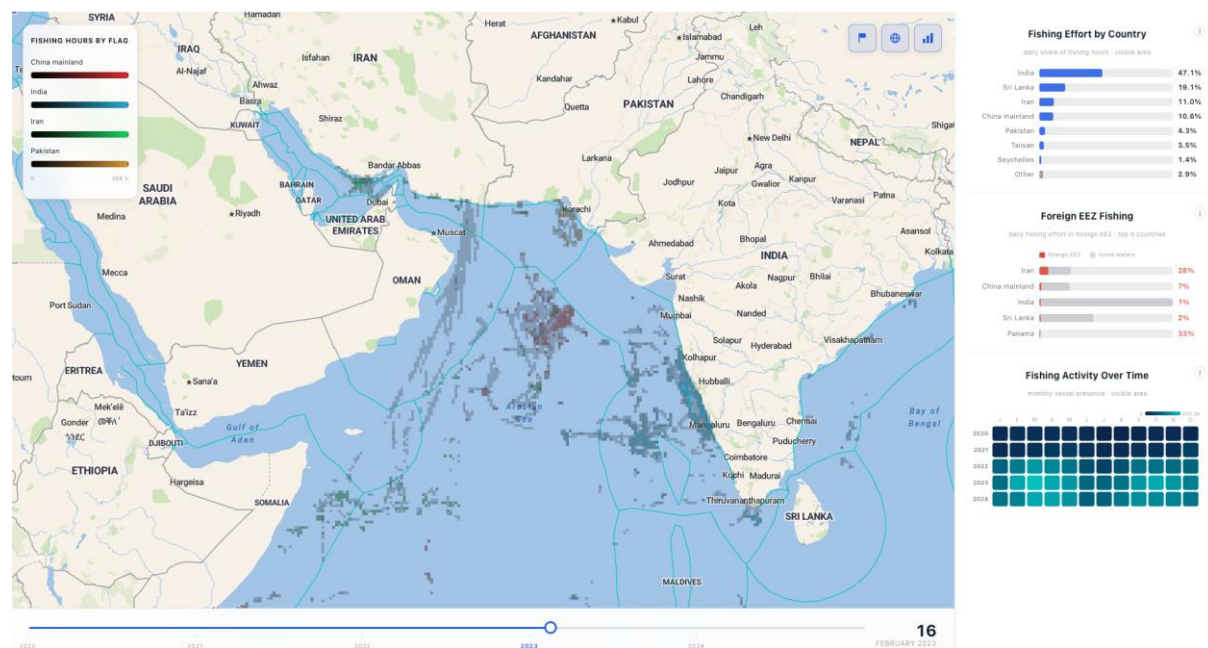


Figure 4 : Use case in Arabian Sea

To illustrate the full potential of *A Fishing Story*, the screenshot above combines several features simultaneously: the country filter is active for China, India, Iran, and Pakistan, the EEZ layer is enabled, and the statistical panel is open, all focused on the Arabian Sea in February 2023.

The contrast between China and India is immediately readable. Indian vessels (blue-green) are concentrated along their own coastline, well within India's EEZ, consistent with a domestic fleet fishing close to home. Chinese vessels (red), by contrast, have no EEZ in this region and are clustered in international waters, far from any sovereign boundary. This pattern is confirmed by the *Fishing Effort by Country* chart: India accounts for 47.1% of fishing hours in the visible area, while China mainland represents only 10.6%, yet spatially, Chinese activity is almost entirely concentrated in the high seas. Indian naval sources confirmed in 2023 that most Chinese vessels are strategically positioned just outside the 200 nautical mile EEZ limit, a behavior our visualization makes directly visible. [Free Press Journal](#)

The *Foreign EEZ Fishing* chart adds another layer: Iran appears fishing 28% of the time in foreign EEZs. Without a bilateral fishing agreement visibly in place, this raises questions that the tool itself cannot answer, but that it invites the user to investigate further. This is precisely the kind of analytical starting point *A Fishing Story* is designed to provide.

Peer assessment:

Nayan Adani:

Nayan led the development of the landing page, including the layout and the transition to the main map view. He implemented the three adaptive statistical charts (*Fishing Effort by Country*, *Foreign EEZ Fishing*, *Fishing Activity Over Time*) as custom SVG components, as well as the country filter feature allowing users to isolate vessels by flag state. He also built the points of interest system, including the zoom behavior and the contextual side panel. As with most features in a collaborative project, these components were iterated upon and refined by the full team over time.

Alexandre Goumaz:

Alexandre was responsible for the data infrastructure and the map at the core of the visualization. He designed and implemented the full data processing pipeline, removing the need for a backend. He also built the main map view, including the data queries and the rendering logic. Finally, he handled the deployment of the website, publishing the processed data on Hugging Face and the static site on GitHub Pages.

Mathieu Senent:

Mathieu designed and implemented the chronological timeline at the bottom of the interface, to navigate day by day through five years of fishing activity. He also led the selection and integration of the base map, including the choice of projection and the adaptive scale that updates with the zoom level. He built the Exclusive Economic Zones (EEZ) layer, which makes maritime borders visible. Finally, Mathieu produced the screencast video presenting the project. As with most features in a collaborative project, these components were iterated upon and refined by the full team over time.